

All Waves Statistical Analyses

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Hierarchical Bayesian Models Cross-Waves analysis

This document runs the models used to analyse the over time changes in perceptions, awareness and experience of open science practices.

The structure of the document follows the Analysis_Plans.txt file.

The variables in the datasets are explained in Meta-Data.txt.

Data Set 2 - Truncated Poisson Regression

Awarweness Score Total

Bayesian regression for how much awareness of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=1000; warmup=500; thin=1;
## post-warmup draws per chain=500, total post-warmup draws=2000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu             1.54     0.00 0.06    1.44    1.50    1.53    1.57    1.66   457
## sigma          0.10     0.01 0.07    0.01    0.04    0.09    0.14    0.27   147
## alpha[1]       1.52     0.00 0.04    1.45    1.50    1.52    1.55    1.60   387
## alpha[2]       1.54     0.00 0.08    1.37    1.49    1.53    1.58    1.71   689
## alpha[3]       1.51     0.00 0.07    1.37    1.47    1.51    1.55    1.64   872
## alpha[4]       1.66     0.01 0.12    1.48    1.56    1.64    1.74    1.91   173
## alpha[5]       1.46     0.01 0.09    1.25    1.40    1.47    1.52    1.61   341
## alpha[6]       1.56     0.00 0.09    1.41    1.50    1.55    1.61    1.76   579
## alpha[7]       1.52     0.00 0.10    1.29    1.47    1.52    1.57    1.72   692
## alpha[8]       1.52     0.00 0.05    1.41    1.48    1.52    1.55    1.62   518
## alpha[9]       1.56     0.00 0.12    1.34    1.49    1.54    1.61    1.86   587
## beta1          0.13     0.00 0.05    0.04    0.10    0.13    0.16    0.23   605
## beta2          0.07     0.00 0.05   -0.04    0.03    0.07    0.11    0.17   605
## beta3          4.40     0.02 0.33    3.73    4.18    4.43    4.64    4.97   359
## lp__          3914.64    0.62 5.65 3905.28 3910.67 3913.93 3917.85 3926.87    82
##               Rhat
## mu             1.01
## sigma          1.00
## alpha[1]       1.01
## alpha[2]       1.00
```

```
## alpha[3] 1.00
## alpha[4] 1.01
## alpha[5] 1.00
## alpha[6] 1.00
## alpha[7] 1.00
## alpha[8] 1.01
## alpha[9] 1.00
## beta1    1.01
## beta2    1.00
## beta3    1.01
## lp__     1.01
##
## Samples were drawn using NUTS(diag_e) at Sun Jan 12 20:57:49 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 767
 Year 1 N = 424
 Year 2 N = 343
 2020 N = 340
 2021 N = 184
 2022 N = 243

Var1	Freq
0	12
1	30
2	41
3	71
4	97
5	85
6	78
7	55
8	298

The output shows that beta credibility intervals are significant for Beta 1.

2.5% = 0.0408976

97.5% = 0.2275188

The interpretation is that students across all waves, who were in their second year of study were more likely to have higher awareness of open science practices compared to students in the first year of study.

The output shows that beta credibility intervals are significant for Beta 3.

2.5% = 3.7290106

97.5% = 4.9731946

The interpretation is that students who answered the survey in 2022 were significantly more likely to have higher awareness of open science practices compared to students who answered the survey in 2020.

177 divergent transitions.

Experience Score Total

Bayesian regression for how much experience of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=1000; warmup=500; thin=1;
## post-warmup draws per chain=500, total post-warmup draws=2000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           0.74     0.00 0.07    0.60    0.69    0.74    0.78    0.87  1035
## sigma        0.14     0.00 0.07    0.05    0.09    0.13    0.18    0.31   617
## alpha[1]     0.73     0.00 0.04    0.64    0.70    0.73    0.76    0.82   896
## alpha[2]     0.73     0.00 0.11    0.49    0.66    0.73    0.79    0.93  1679
## alpha[3]     0.73     0.00 0.08    0.56    0.67    0.73    0.79    0.89  1907
## alpha[4]     0.82     0.00 0.11    0.62    0.74    0.81    0.89    1.04  1146
## alpha[5]     0.65     0.00 0.12    0.37    0.57    0.66    0.73    0.86  1131
## alpha[6]     0.79     0.00 0.07    0.65    0.74    0.79    0.84    0.94   725
## alpha[7]     0.60     0.01 0.16    0.20    0.51    0.63    0.72    0.86   977
## alpha[8]     0.91     0.00 0.07    0.77    0.86    0.91    0.95    1.04   718
## alpha[9]     0.67     0.00 0.08    0.51    0.62    0.67    0.73    0.84   779
## beta1        0.26     0.00 0.04    0.18    0.23    0.26    0.29    0.34  1426
## beta2       -0.39     0.00 0.07   -0.53   -0.44   -0.39   -0.34   -0.25   952
## beta3       -0.02     0.00 0.06   -0.14   -0.07   -0.02    0.02    0.10   655
## lp__       -359.82     0.13 3.15 -366.54 -361.84 -359.63 -357.64 -354.22   577
##               Rhat
## mu           1.00
## sigma        1.00
## alpha[1]     1.01
## alpha[2]     1.00
## alpha[3]     1.00
## alpha[4]     1.00
## alpha[5]     1.00
## alpha[6]     1.00
## alpha[7]     1.00
## alpha[8]     1.01
## alpha[9]     1.00
## beta1        1.00
## beta2        1.01
## beta3        1.01
## lp__         1.01
##
## Samples were drawn using NUTS(diag_e) at Sun Jan 12 20:58:51 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 987

Year 1 N = 571

Year 2 N = 416

2020 N = 340

2021 N = 184

2022 N = 463

Var1	Freq
0	211
1	167
2	186
3	178
4	126
5	75
6	28
7	11
8	5

The output shows that beta credibility intervals are significant for Beta 1.

2.5% = 0.1765044

97.5% = 0.3446366

The interpretation is that students across all waves, who were in their second year of study were more likely to have higher experience of open science practices compared to students in the first year of study.

The output shows that beta credibility intervals are significant for Beta 2 in the negative direction.

2.5% = -0.5327447

97.5% = -0.2475686

The interpretation is that students who answered the survey in 2020 were significantly more likely to have higher experience of open science practices compared to students who answered the survey in 2022.

Data Set 2 - Linear Regression

Conceptual Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4.

The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=1000; warmup=500; thin=1;
## post-warmup draws per chain=500, total post-warmup draws=2000.
##
##               mean se_mean   sd  2.5%   25%   50%   75%  97.5% n_eff Rhat
## mu            0.59     0.00 0.05   0.50   0.56   0.59   0.62   0.69  1501 1.00
## sigma         0.38     0.00 0.01   0.36   0.38   0.38   0.39   0.40  1991 1.00
## sigma_a       0.12     0.00 0.04   0.06   0.08   0.11   0.14   0.22  1077 1.00
## alpha[1]      0.55     0.00 0.03   0.50   0.53   0.55   0.57   0.60  1439 1.00
## alpha[2]      0.53     0.00 0.07   0.38   0.48   0.53   0.57   0.66  2173 1.00
## alpha[3]      0.61     0.00 0.06   0.50   0.58   0.62   0.65   0.72  1796 1.00
## alpha[4]      0.70     0.00 0.07   0.57   0.65   0.70   0.75   0.84  2034 1.00
## alpha[5]      0.49     0.00 0.07   0.35   0.44   0.49   0.54   0.62  1915 1.00
## alpha[6]      0.70     0.00 0.04   0.61   0.67   0.70   0.73   0.79  1142 1.00
## alpha[7]      0.58     0.00 0.09   0.40   0.53   0.58   0.64   0.77  1911 1.00
## alpha[8]      0.66     0.00 0.04   0.59   0.63   0.66   0.69   0.73  1165 1.00
## alpha[9]      0.50     0.00 0.05   0.42   0.47   0.50   0.53   0.59  1223 1.00
```

```
## beta1      0.05      0.00 0.03      0.00      0.03      0.05      0.06      0.10      2050 1.00
## beta2     -0.07      0.00 0.04     -0.15     -0.09     -0.06     -0.04      0.01      1371 1.00
## beta3      0.01      0.00 0.04     -0.06     -0.01      0.01      0.04      0.09      1083 1.00
## lp__       471.31      0.11 2.93    464.77    469.48    471.61    473.43    476.09      661 1.01
##
## Samples were drawn using NUTS(diag_e) at Sun Jan 12 21:04:05 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 983
 Year 1 N = 569
 Year 2 N = 414
 2020 N = 337
 2021 N = 183
 2022 N = 463

The output shows that all beta credibility intervals cross 0.

Situational Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4.

The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=1000; warmup=500; thin=1;
## post-warmup draws per chain=500, total post-warmup draws=2000.
##
##          mean se_mean   sd  2.5%   25%   50%   75%  97.5% n_eff Rhat
## mu          0.40     0.00 0.06   0.27   0.36   0.40   0.44   0.52  1519 1.00
## sigma        0.49     0.00 0.01   0.47   0.48   0.49   0.50   0.52  2425 1.00
## sigma_a      0.14     0.00 0.06   0.07   0.10   0.13   0.17   0.29  1030 1.00
## alpha[1]     0.29     0.00 0.03   0.23   0.27   0.29   0.32   0.36  1552 1.00
## alpha[2]     0.36     0.00 0.09   0.17   0.30   0.36   0.42   0.53  2348 1.00
## alpha[3]     0.42     0.00 0.07   0.29   0.38   0.42   0.47   0.57  1781 1.00
## alpha[4]     0.51     0.00 0.09   0.35   0.45   0.51   0.57   0.70  2038 1.00
## alpha[5]     0.27     0.00 0.10   0.06   0.20   0.27   0.34   0.46  2001 1.00
## alpha[6]     0.53     0.00 0.06   0.41   0.49   0.53   0.57   0.64  1222 1.01
## alpha[7]     0.42     0.00 0.11   0.20   0.35   0.42   0.50   0.65  1954 1.00
## alpha[8]     0.47     0.00 0.05   0.38   0.44   0.47   0.50   0.57  1155 1.00
## alpha[9]     0.30     0.00 0.06   0.17   0.26   0.30   0.34   0.42  1338 1.00
## beta1        0.10     0.00 0.03   0.04   0.08   0.10   0.12   0.17  2389 1.00
## beta2       -0.08     0.00 0.05  -0.18  -0.11  -0.07  -0.04   0.02  1557 1.01
## beta3       -0.04     0.00 0.05  -0.14  -0.08  -0.04  -0.01   0.05  1080 1.01
## lp__        218.60     0.12 3.09   211.35   216.73   219.03   220.91   223.35   617 1.00
##
## Samples were drawn using NUTS(diag_e) at Sun Jan 12 21:04:54 2025.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 976
Year 1 N = 565
Year 2 N = 411
2020 N = 330
2021 N = 183
2022 N = 463

The output shows that beta credibility intervals do not cross 0 only for the Beta 1.

2.5% = 0.0395106

97.5% = 0.1660442

The interpretation is that students across all waves, who were in their second year of study were more likely to have more positive situational perceptions of open science practices compared to students in the first year of study.